

# Topic 3

## Dynamics

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Revision Booklet

**This booklet covers:**

- Mass, Force and Newton's Laws of Motion
- Linear Momentum and Impulse
- Non-Uniform Motion and Terminal Velocity
- Conservation of Momentum
- Elastic and Inelastic Collisions

## Mass, Force and Newton's Laws of Motion

### Mass

**Mass** is the property of an object that resists change in motion (inertia). It is a scalar quantity measured in kilograms (kg). A larger mass requires a larger force to produce the same acceleration.

### Newton's First Law

An object remains at rest or continues to move with **constant velocity** unless acted upon by a **resultant (net) force**.

Equivalently: if the resultant force on an object is zero, its acceleration is zero.

### Newton's Second Law

The **resultant force** on an object is directly proportional to its **rate of change of momentum**, and acts in the same direction as the rate of change of momentum:

$$F = \frac{\Delta p}{\Delta t} = \frac{\Delta(mv)}{\Delta t}$$

For constant mass this simplifies to:

$$F = ma$$

- $F$ : resultant force (N);  $m$ : mass (kg);  $a$ : acceleration ( $\text{m s}^{-2}$ ).
- The resultant force and acceleration are always in the **same direction**.
- 1 N: the force that gives a mass of 1 kg an acceleration of  $1 \text{ m s}^{-2}$ .

### Newton's Third Law

If object A exerts a force on object B, then object B exerts an **equal and opposite force** on object A.

These forces are:

- Equal in magnitude.
- Opposite in direction.
- Of the **same type** (e.g. both gravitational, both contact).
- Acting on **different objects** — they never cancel because they act on different bodies.

### Newton's Third Law Pairs

A common error is to confuse Newton's third law pairs with forces in equilibrium. Two forces balance (sum to zero) when they act on the *same* object; Newton's third law pairs act on *different* objects. For example: weight ( $mg$  downward on book) and normal contact force ( $N$  upward on book) are *not* a Newton's third law pair — they are equilibrium forces on the same object. The Newton's third law pair of the book's weight is the gravitational

pull the book exerts on the Earth.

### Weight

The **weight** of an object is the gravitational force acting on it due to a gravitational field:

$$W = mg$$

- $g = 9.81 \text{ m s}^{-2}$ : gravitational field strength / acceleration of free fall.
- Weight acts at the object's **centre of gravity**.
- Weight is a vector (downward); mass is a scalar.

## Linear Momentum and Impulse

### Linear Momentum

The **linear momentum** of an object is the product of its mass and velocity:

$$p = mv$$

- Unit:  $\text{kg m s}^{-1}$  (or  $\text{N s}$ ).
- Momentum is a **vector** — direction matters.
- Always state or define the positive direction when solving problems.

### Force as Rate of Change of Momentum

Newton's second law in its most general form:

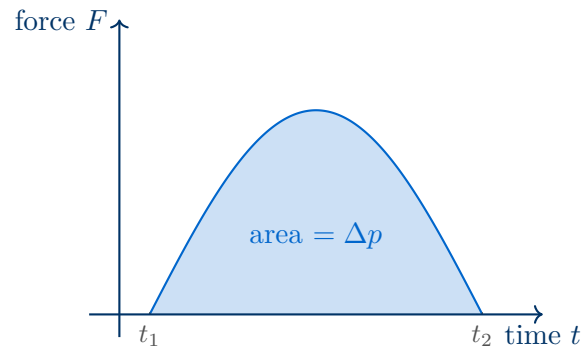
$$F = \frac{\Delta p}{\Delta t}$$

Rearranging:  $F \Delta t = \Delta p$

The product  $F \Delta t$  is called the **impulse**:

$$\text{Impulse} = F \Delta t = \Delta p = mv - mu$$

- Unit:  $\text{N s} \equiv \text{kg m s}^{-1}$ .
- Impulse equals the **change in momentum**.
- For a variable force, impulse = area under a force–time graph.



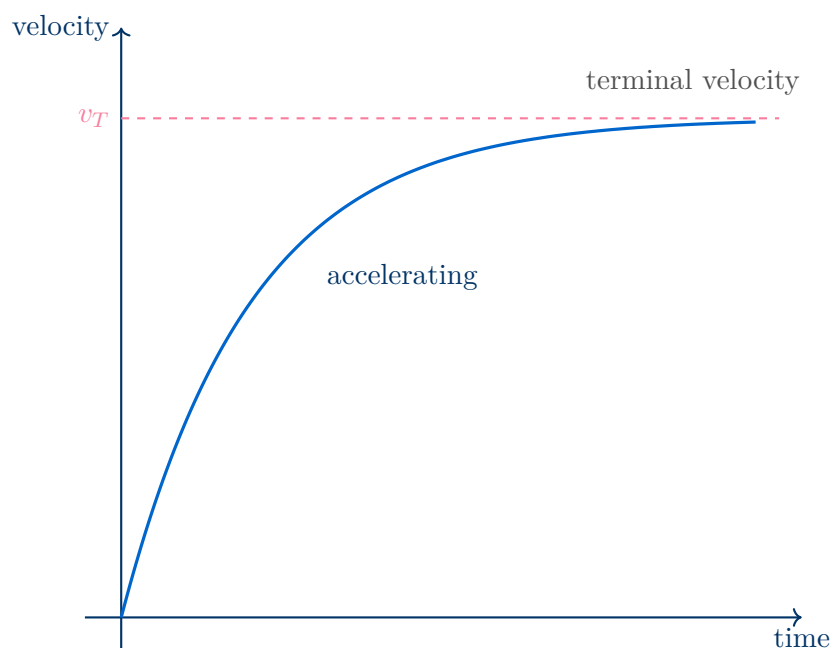
## Non-Uniform Motion and Terminal Velocity

### Friction and Drag Forces

- **Friction:** contact force opposing relative motion between surfaces.
- **Viscous drag / air resistance:** resistive force on an object moving through a fluid. A simple model: drag force **increases as speed increases**.
- Both forces act opposite to the direction of motion.

### Terminal Velocity

An object falling through a fluid **reaches terminal velocity** when the **drag force equals the driving force** (weight for a falling object). At this point the resultant force is zero and acceleration is zero — the object moves at constant velocity.

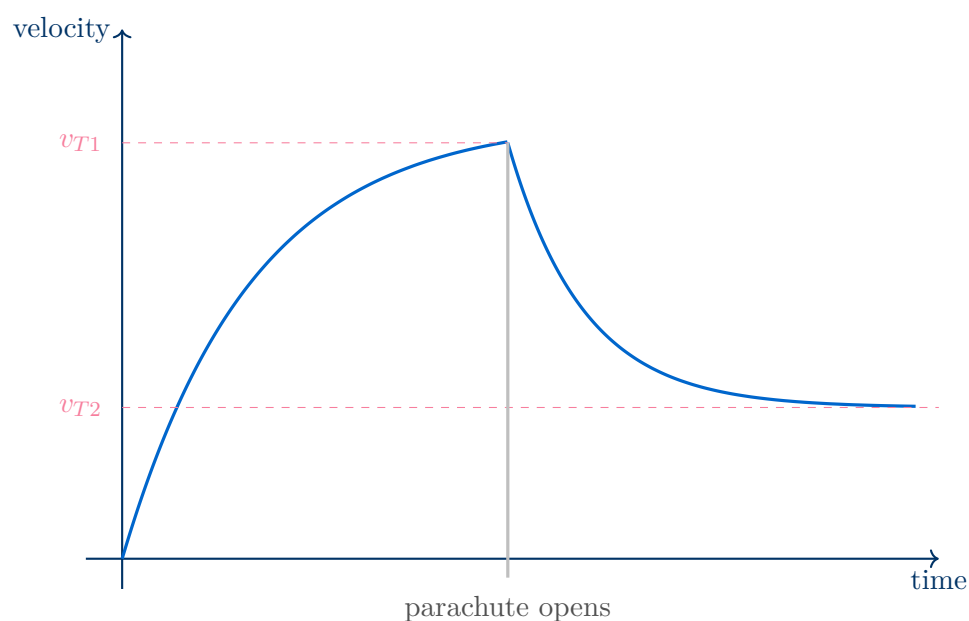


**Stages of Free Fall with Air Resistance**

1. **Initially:** velocity = 0, drag = 0. Resultant force =  $mg$  downward. Acceleration =  $g$ .
2. **As speed increases:** drag increases. Resultant force decreases. Acceleration decreases (gradient of  $v$ - $t$  graph decreases).
3. **At terminal velocity  $v_T$ :** drag =  $mg$ . Resultant force = 0. Acceleration = 0. Constant velocity.

For a parachutist who opens their parachute after reaching terminal velocity:

- Drag suddenly increases greatly  $\Rightarrow$  resultant force is now *upward*  $\Rightarrow$  deceleration.
- Speed decreases  $\Rightarrow$  drag decreases until a new, lower terminal velocity is reached.



## Conservation of Linear Momentum

### Principle of Conservation of Momentum

The **total linear momentum** of a system of objects remains constant provided **no external resultant force** acts on the system.

$$\sum p_{\text{before}} = \sum p_{\text{after}}$$

This follows directly from Newton's second and third laws.

### Applying Conservation of Momentum

1. Define a **positive direction**.
2. Write down total momentum **before** the collision/explosion.
3. Write down total momentum **after**, using + or – for direction.
4. Set before = after and solve.

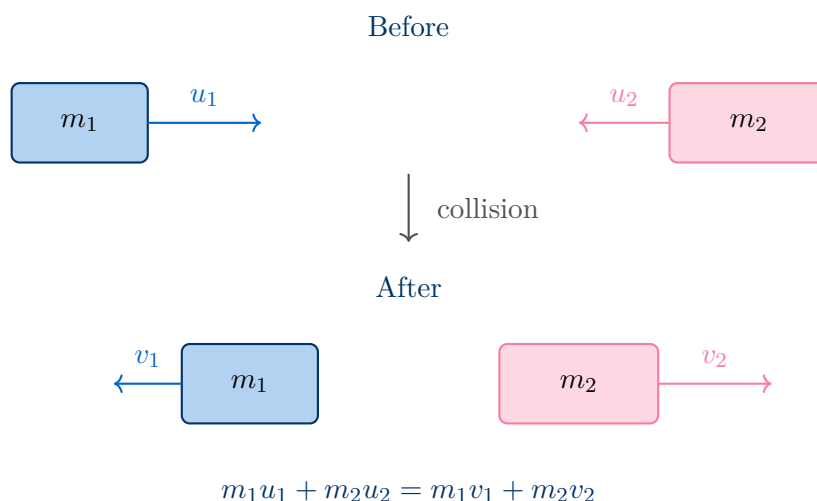
For a two-body collision:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

For an explosion starting from rest (total initial momentum = 0):

$$0 = m_1 v_1 + m_2 v_2 \implies m_1 v_1 = -m_2 v_2$$

The two objects move in **opposite directions**.



## Elastic and Inelastic Collisions

### Elastic Collision

In an **elastic collision**:

- Total **momentum** is conserved (always).
- Total **kinetic energy** is conserved.
- The **relative speed of approach** equals the **relative speed of separation**:

$$u_1 - u_2 = v_2 - v_1$$

Perfectly elastic collisions are rare in practice (e.g. collisions between gas molecules, some subatomic particles).

### Inelastic Collision

In an **inelastic collision**:

- Total **momentum** is conserved (always).
- Total **kinetic energy is not conserved** — some is converted to heat, sound, or deformation.
- In a **perfectly inelastic** collision, the objects **stick together** and move as one — maximum kinetic energy is lost (but momentum is still conserved).

|                                       | Elastic | Inelastic                      |
|---------------------------------------|---------|--------------------------------|
| Momentum conserved                    | Yes     | Yes                            |
| Kinetic energy conserved              | Yes     | No                             |
| Relative speed: approach = separation | Yes     | No                             |
| Objects stick together                | No      | Possibly (perfectly inelastic) |

### Checking for Elastic Collision

To determine whether a collision is elastic:

1. Apply conservation of momentum to find unknown velocities.
2. Calculate total KE before:  $\sum \frac{1}{2}mv^2$  (using initial speeds).
3. Calculate total KE after:  $\sum \frac{1}{2}mv^2$  (using final speeds).
4. If KE before = KE after  $\Rightarrow$  **elastic**. If KE is lost  $\Rightarrow$  **inelastic**.
5. Alternatively, check if  $u_1 - u_2 = v_2 - v_1$ .

## Formula Summary Sheet

| Formula                                          | Quantity                            | Units                |
|--------------------------------------------------|-------------------------------------|----------------------|
| $F = ma$                                         | Newton's second law (constant mass) | N                    |
| $W = mg$                                         | Weight                              | N                    |
| $p = mv$                                         | Linear momentum                     | kg m s <sup>-1</sup> |
| $F = \Delta p / \Delta t$                        | Force as rate of change of momentum | N                    |
| $F \Delta t = \Delta p$                          | Impulse                             | N s                  |
| $\sum p_{\text{before}} = \sum p_{\text{after}}$ | Conservation of momentum            | kg m s <sup>-1</sup> |
| $u_1 - u_2 = v_2 - v_1$                          | Elastic collision condition         | m s <sup>-1</sup>    |

**Newton's laws word-for-word:**

**1st:** An object remains at rest or moves at constant velocity unless a resultant force acts.

**2nd:** Resultant force equals rate of change of momentum ( $F = \Delta p / \Delta t$ ).

**3rd:** For every action there is an equal and opposite reaction (on a different object, same type of force).

**Momentum is always conserved** in all collisions (elastic and inelastic).

**Kinetic energy is only conserved** in elastic collisions.

## Worked Examples

### Example 1 — Newton's Second Law

**Question:** A car of mass 1200 kg accelerates from  $8.0 \text{ m s}^{-1}$  to  $20 \text{ m s}^{-1}$  in 6.0 s. The total resistive force is 400 N. Calculate (a) the acceleration, (b) the driving force.

#### Solution

$$(a) \ a = \frac{v - u}{t} = \frac{20 - 8.0}{6.0} = \mathbf{2.0 \text{ m s}^{-2}}$$

$$(b) \ F_{\text{net}} = ma = 1200 \times 2.0 = 2400 \text{ N}$$

$$F_{\text{drive}} - F_{\text{resist}} = F_{\text{net}} \implies F_{\text{drive}} = 2400 + 400 = \mathbf{2800 \text{ N}}$$

### Example 2 — Impulse and Change in Momentum

**Question:** A ball of mass 0.40 kg travelling at  $12 \text{ m s}^{-1}$  hits a wall and rebounds at  $9.0 \text{ m s}^{-1}$ . The collision lasts 0.025 s. Calculate (a) the impulse, (b) the average force on the ball.

#### Solution

Taking towards the wall as positive:

$$(a) \ \Delta p = mv - mu = 0.40 \times (-9.0) - 0.40 \times 12 = -3.6 - 4.8 = \mathbf{-8.4 \text{ N s}}$$

Magnitude of impulse = 8.4 N s (directed away from wall).

$$(b) \ F = \Delta p / \Delta t = -8.4 / 0.025 = \mathbf{-336 \text{ N}}$$

The average force on the ball is 336 N directed away from the wall.

### Example 3 — Conservation of Momentum

**Question:** Trolley A (mass 2.0 kg, velocity  $+4.0 \text{ m s}^{-1}$ ) collides with stationary trolley B (mass 3.0 kg). After the collision, A moves at  $+1.0 \text{ m s}^{-1}$ . Find the velocity of B and determine whether the collision is elastic.

#### Solution

**Conservation of momentum** (taking right as positive):

$$p_{\text{before}} = 2.0 \times 4.0 + 3.0 \times 0 = 8.0 \text{ kg m s}^{-1}$$

$$p_{\text{after}} = 2.0 \times 1.0 + 3.0 \times v_B = 2.0 + 3.0v_B$$

$$8.0 = 2.0 + 3.0v_B \implies v_B = \mathbf{2.0 \text{ m s}^{-1}}$$

**Check for elastic collision:**

$$\text{KE before} = \frac{1}{2}(2.0)(4.0)^2 + 0 = 16.0 \text{ J}$$

$$\text{KE after} = \frac{1}{2}(2.0)(1.0)^2 + \frac{1}{2}(3.0)(2.0)^2 = 1.0 + 6.0 = 7.0 \text{ J}$$

KE is not conserved ( $16.0 \neq 7.0$ )  $\implies$  the collision is **inelastic**.

**Example 4 — Perfectly Inelastic Collision and Explosion**

**Question:** (a) A bullet of mass 0.020 kg travelling at  $350 \text{ m s}^{-1}$  embeds in a stationary block of mass 1.98 kg. Find their common velocity after impact.

(b) A stationary rocket of mass 800 kg explodes into two fragments. Fragment A (300 kg) moves at  $+120 \text{ m s}^{-1}$ . Find the velocity of fragment B.

**Solution**

(a) Perfectly inelastic — objects stick together:

$$mu = (m + M)v \implies v = \frac{0.020 \times 350}{0.020 + 1.98} = \frac{7.0}{2.0} = \mathbf{3.5 \text{ m s}^{-1}}$$

(b) Total initial momentum = 0 (at rest):

$$0 = 300 \times 120 + 500 \times v_B \implies v_B = \frac{-36000}{500} = \mathbf{-72 \text{ m s}^{-1}}$$

Fragment B moves at  $72 \text{ m s}^{-1}$  in the opposite direction to A.

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** State Newton's three laws of motion. For each law, give a practical example.

*[6 marks]*

**Q2.** Define linear momentum and state its unit. Explain what is meant by the impulse of a force and state how it relates to momentum.

*[4 marks]*

**Q3.** A skydiver of mass 75 kg (including equipment) falls from rest. Describe and explain the motion from the moment of jumping until a terminal velocity of  $55 \text{ m s}^{-1}$  is reached. Include reference to the forces acting throughout.

*[5 marks]*

**Q4.** Distinguish between an elastic and an inelastic collision. State which quantities are conserved in each.

*[3 marks]*

### Section B — Longer Structured Questions

**Q5.** A force–time graph for a golf club striking a ball (mass 0.046 kg, initially at rest) shows a peak force of 2400 N over a contact time of  $5.0 \times 10^{-4}$  s. Assume the force–time graph is a triangle.

(a) Calculate the impulse given to the ball.

*[2 marks]*

(b) Hence calculate the speed of the ball immediately after impact.

*[2 marks]*

(c) Calculate the average force on the club from the ball during contact, and state its direction.

*[2 marks]*

**Q6.** Two ice skaters, A (mass 60 kg) and B (mass 80 kg), stand at rest facing each other on a frictionless ice rink. They push off from each other. After the push, skater A moves at  $2.4 \text{ m s}^{-1}$  to the left.

- (a) Calculate the velocity of skater B after the push.

*[3 marks]*

- (b) Calculate the total kinetic energy after the push. Explain whether this is consistent with conservation of energy.

*[3 marks]*

**Q7.** Ball P (mass 0.30 kg, velocity  $+6.0 \text{ m s}^{-1}$ ) collides head-on with ball Q (mass 0.50 kg, velocity  $-2.0 \text{ m s}^{-1}$ ). After the collision, ball P has velocity  $-1.5 \text{ m s}^{-1}$ .

- (a) Use conservation of momentum to find the velocity of Q after the collision.

*[3 marks]*

- (b) Calculate the kinetic energy before and after the collision and determine whether the collision is elastic or inelastic.

*[3 marks]*

- (c) Verify your answer to (a) using the elastic collision condition  $u_1 - u_2 = v_2 - v_1$ .

*[2 marks]*

## Mark Scheme and Answers

**Q1.** 1st: object at rest/constant velocity unless resultant force acts [1]; e.g. book on table [1]. 2nd: resultant force = rate of change of momentum /  $F = ma$  [1]; e.g. pushing a trolley [1]. 3rd: equal and opposite forces on different objects, same type [1]; e.g. swimmer pushing wall, wall pushes swimmer back [1].

**Q2.** Momentum = mass  $\times$  velocity [1]; unit:  $\text{kg m s}^{-1}$  or  $\text{N s}$  [1]. Impulse = force  $\times$  time =  $F\Delta t$  [1]; impulse equals the change in momentum [1].

**Q3.** Initially only weight acts; acceleration =  $g = 9.81 \text{ m s}^{-2}$  downward [1]. As speed increases, air resistance increases [1]; resultant force decreases, acceleration decreases [1]; gradient of  $v-t$  graph decreases [1]; at terminal velocity ( $55 \text{ m s}^{-1}$ ): drag =  $mg = 75 \times 9.81 = 736 \text{ N}$ , resultant = 0, acceleration = 0 [1].

**Q4.** Elastic: both momentum and kinetic energy conserved [1]. Inelastic: momentum conserved, kinetic energy not conserved (converted to heat/sound/deformation) [1]. In all collisions, momentum is always conserved [1].

**Q5(a).** Triangular area: impulse =  $\frac{1}{2} \times F_{\text{max}} \times t = \frac{1}{2} \times 2400 \times 5.0 \times 10^{-4} = \mathbf{0.60 \text{ N s}}$  [2].

**Q5(b).**  $v = \Delta p/m = 0.60/0.046 = \mathbf{13 \text{ m s}^{-1}}$  [2].

**Q5(c).** By Newton's 3rd law, force on club =  $0.60/5.0 \times 10^{-4} = \mathbf{1200 \text{ N}}$  [1]; directed opposite to ball's motion (back toward the golfer) [1].

**Q6(a).** Initial total momentum = 0;  $0 = 60 \times (-2.4) + 80 \times v_B$  [1];  $80v_B = 144$  [1];  $v_B = +\mathbf{1.8 \text{ m s}^{-1}}$  (to the right) [1].

**Q6(b).**  $\text{KE} = \frac{1}{2}(60)(2.4)^2 + \frac{1}{2}(80)(1.8)^2 = 172.8 + 129.6 = \mathbf{302 \text{ J}}$  [2]. Initial KE = 0; energy is not conserved — the skaters' muscles do work (chemical energy converted to kinetic energy); this is an explosion, not a collision, so KE need not be conserved [1].

**Q7(a).**  $p_{\text{before}} = 0.30 \times 6.0 + 0.50 \times (-2.0) = 1.8 - 1.0 = 0.80 \text{ kg m s}^{-1}$  [1];  $p_{\text{after}} = 0.30 \times (-1.5) + 0.50 \times v_Q = -0.45 + 0.50v_Q$  [1];  $0.80 = -0.45 + 0.50v_Q \Rightarrow v_Q = +\mathbf{2.5 \text{ m s}^{-1}}$  [1].

**Q7(b).** KE before =  $\frac{1}{2}(0.30)(6.0)^2 + \frac{1}{2}(0.50)(2.0)^2 = 5.4 + 1.0 = 6.4 \text{ J}$  [1]; KE after =  $\frac{1}{2}(0.30)(1.5)^2 + \frac{1}{2}(0.50)(2.5)^2 = 0.3375 + 1.5625 = 1.9 \text{ J}$  [1]; KE not conserved  $\Rightarrow$  **inelastic** [1].

**Q7(c).**  $u_1 - u_2 = 6.0 - (-2.0) = 8.0 \text{ m s}^{-1}$ ;  $v_2 - v_1 = 2.5 - (-1.5) = 4.0 \text{ m s}^{-1}$  [1];  $8.0 \neq 4.0 \Rightarrow$  confirms the collision is **inelastic** (elastic condition not satisfied) [1].

## Revision Checklist

| Learning Objective                                                                                               | Confidence (1–3) |
|------------------------------------------------------------------------------------------------------------------|------------------|
| <input type="checkbox"/> Understand that mass is the property resisting change in motion                         |                  |
| <input type="checkbox"/> State and apply Newton's first law                                                      |                  |
| <input type="checkbox"/> Use $F = ma$ and $F = \Delta p / \Delta t$ ; know they are equivalent for constant mass |                  |
| <input type="checkbox"/> State and apply Newton's third law; identify Newton's third law pairs correctly         |                  |
| <input type="checkbox"/> Define weight as $W = mg$ and locate it at the centre of gravity                        |                  |
| <input type="checkbox"/> Define momentum as $p = mv$ and state its unit                                          |                  |
| <input type="checkbox"/> Define impulse as $F\Delta t = \Delta p$ ; find impulse from area under $F-t$ graph     |                  |
| <input type="checkbox"/> Describe qualitatively frictional and drag forces                                       |                  |
| <input type="checkbox"/> Explain terminal velocity in terms of forces and draw $v-t$ graph                       |                  |
| <input type="checkbox"/> Describe the motion of a parachutist opening their parachute                            |                  |
| <input type="checkbox"/> State the principle of conservation of momentum                                         |                  |
| <input type="checkbox"/> Apply conservation of momentum to 1D and 2D collisions and explosions                   |                  |
| <input type="checkbox"/> Distinguish elastic and inelastic collisions by KE conservation                         |                  |
| <input type="checkbox"/> Use $u_1 - u_2 = v_2 - v_1$ to identify/verify an elastic collision                     |                  |
| <input type="checkbox"/> Understand that momentum is always conserved; KE only in elastic collisions             |                  |
| Key: 1 = Need more work    2 = Getting there    3 = Confident                                                    |                  |

## Good luck with your revision!

Dynamics is Newton's three laws working together. Everything from terminal velocity to collisions follows from  $F = \Delta p / \Delta t$  and the simple rule that momentum is always conserved.

Get those two ideas solid and you have the whole topic.